



Answers & Solutions:

1. **A. decreases**
2. **B. 10 MΩ**
3. **B. Controlling device**
4. **C. Eddy current**
5. **A. Electromagnetic deflection.**
6. **A. oscilloscope**
7. **A. the amount of current needed in the moving-coil to produce full scale deflection of the meter's pointer. For a moving-coil meter movement, I_M is**
8. **D. A square wave normally contains the fundamental and even harmonics**
9. **A. 271 V**

Solution:

$$R_v = S(V_{\text{scale}})$$

$$R_v = (20,000 \Omega/V)(300 V)$$

$$R_v = 6 M\Omega$$

$$I_v = \frac{250 V}{6 M\Omega}$$

$$V_u = I_v(500 k\Omega + 6 M\Omega)$$

$$V_u = \left(\frac{250 V}{6 M\Omega}\right)(500 k\Omega + 6 M\Omega)$$

$$V_u = 271 V \rightarrow \text{Ans.}$$

10. **A. the power in the circuit being tested must be off.**

11. **A. 0.02004 Ω**

Solution:

$$R_{SH} = \frac{I_m R_m}{I - I_m}$$

$$R_{SH} = \frac{(10 \text{ mA})(10 \Omega)}{5 \text{ A} - 10 \text{ mA}}$$

$$R_{SH} = 0.02004 \Omega \rightarrow \text{Ans.}$$

12. **B. multiplier**

13. **B. increases.**

14. **C. It allows for measurement of a wide range of currents.**

15. **B. use the volts/division control to adjust the amplitude of the signal before it is applied to the vertical deflection plates.**

16. **C. increase**

17. **B. The analog VDM.**

18. **A. 0.27 mA – 0.33 mA**

Solution:

$$0.3 \text{ mA}(3\%)$$

$$1 \text{ mA}(0.03) = 0.03$$

$$0.3 \text{ mA} - 0.03 = 0.27 \text{ mA} \rightarrow \text{Ans.}$$

$$0.3 \text{ mA} + 0.03 = 0.33 \text{ mA} \rightarrow \text{Ans.}$$

19. **A. Attraction type**

20. **D. 98.5 kΩ**

Solution:

$$V = I_{FS}(R_s + R_m)$$

$$15 \text{ V} = 150 \mu\text{A}(R_s + 1.5 k\Omega)$$

$$R_s = 98.5 k\Omega \rightarrow \text{Ans.}$$

21. **C. the meter is read when the pointer and its mirror reflection appear as one.**

22. **B. An even scale, measuring a.c.**

23. **C. It can detect ac voltages.**

24. **A. 1.96**

Solution:

$$\% \text{error} = \frac{1 k\Omega}{50 k\Omega + 1 k\Omega} \times 100\%$$

$$\% \text{error} = 1.96\% \rightarrow \text{Ans.}$$

25. **B. An amp-clamp probe**

26. **B. analog oscilloscope**

27. **C. 9000**

Solution:

$$\alpha_{ac} = 0.9\alpha_{dc}$$

$$\alpha_{ac} = 0.9(10000)$$

$$\alpha_{ac} = 9000 \rightarrow \text{Ans.}$$

where:

$$\alpha_{dc} = \frac{1}{I_{FS}} = \frac{1}{100 \mu\text{A}}$$

$$\alpha_{dc} = 10000$$

28. **B. Air**

29. **D. Is placed in parallel with the circuit that works from the supply.**

30. **C. current measurements**

31. **A. voltmeter**

32. **C. A digital voltmeter requires a D/A converter**

33. **D. The electromagnet meter is more rugged.**

34. **D. All choices are correct**

35. **B. very low internal resistance.**

36. **B. repulsion type**

37. **D. Both A and B.**

38. **C. 359 kΩ**

Solution:

$$R_s = \frac{V_{DC}}{I_{FS}} - R_m = \frac{0.45 V_{\text{rmsin}}}{I_{FS}} - R_m$$

$$R_s = \frac{0.45(40 V_{\text{RMS}})}{50 \mu\text{A}} - 1000 \Omega$$

$$R_s = 359 k\Omega \rightarrow \text{Ans.}$$

39. **D. An even scale, measuring d.c**

40. **C. shunt**

41. **D. Both A and C**

42. **D. high-resistance circuits.**

43. **B. 4.76 %**



Solution:

$$\%error = \frac{R_m}{R_m + R_L} \times 100\%$$

$$\%error = \frac{50 \Omega}{50 \Omega + 1 \text{ k}\Omega} \times 100\%$$

$$\%error = 4.76\% \rightarrow \text{Ans.}$$

44. C. Double beam oscilloscope

45. D. Energy

46. D. Hairspring

47. C. very high internal resistance.

48. B. 14 A

49. B. A series-value resistance

50. C. ammeter

END